SESSION 04

Welcome Back!

Intro to Supervised Learning

Types of ML · Regression · Classification · Vocabulary

AI MASTER CLASS

What We'll Learn Today

01 **Types of Machine Learning**

Supervised vs Unsupervised — what makes them different

02 **Supervised Learning Overview**

The full pipeline: data → model → predictions

03 **Regression**

When the answer is a number — predicting continuous values

04 **Classification**

When the answer is a category — spam, not spam, species…

05 **Machine Learning Vocabulary**

Target, Features, Example, Label — the language of ML

06 **The Iris Dataset Deep Dive**

Walk through a real dataset and identify all vocabulary

The Big Picture — What is Machine Learning?

Machine Learning: A way to teach computers to learn from data — without being explicitly programmed for every scenario.

Traditional Programming vs Machine Learning

Traditional Programming

Input + Rules → Output

You write every rule manually:

```
IF income > 50k AND age > 30
  THEN approve loan
```

Problem: impossible to write rules for complex tasks like recognising faces or understanding speech.

Machine Learning

Input + Output → Rules (learned!)

You feed examples, the algorithm discovers patterns:

```
1000 approved loans + features
→ model learns rules by itself
```

Works for complex tasks — images, text, fraud detection, medical diagnosis, and much more!

Types of Machine Learning

The single most important question: Do we have LABELS (correct answers) in our data?

SUPERVISED LEARNING

Data has KNOWN outcomes
Each training example has a label — the correct answer is already known.
Examples:
- Email labeled spam / not spam
- House with known sale price
- Patient with confirmed diagnosis
- Photo labeled 'cat' or 'dog'

Why: We want the model to learn to predict these labels for NEW, unseen data.

VS

UNSUPERVISED LEARNING

Data has UNKNOWN outcomes
No labels. The algorithm must discover hidden structure or patterns on its own.
Examples:
- Group customers by behaviour
- Detect unusual network traffic
- Compress images (find structure)
- Recommend products

Why: Many real-world problems have no labels — labeling data is expensive and slow.

Supervised vs Unsupervised — Real-Life Analogies

🎓 Supervised Learning — Learning with a teacher

Scenario: A student studies past exam papers that include the ANSWERS. They learn: 'This type of question → this type of answer.'

In ML: The model learns from labeled data (questions = features, answers = labels).

🔍 Unsupervised Learning — Exploring without a guide

Scenario: A tourist arrives in a city with no map. They walk around and naturally notice: there's a shopping area, a quiet residential zone, a busy market zone — they discover structure themselves.

In ML: The algorithm finds natural groups or patterns in data with no labels provided.

Key insight: Supervised learning is more common in practice — it's more accurate when labels are available. This entire session focuses on it.

Supervised Learning — The Full Pipeline

Every supervised learning project follows the same 3-stage pipeline:

1 Training Phase

Labeled Training Data

Data points WITH known answers
(e.g. 1,000 emails labeled
spam / not spam)

→

MODEL

→

MODEL

Algorithm learns patterns
from the labeled examples
and stores them internally

2 Prediction Phase

New Unlabeled Data

Fresh data points with
UNKNOWN labels
(e.g. a brand new email)

→

MODEL

→

PREDICTIONS

The trained model applies
its learned patterns to
predict the label

The key insight: training and prediction are SEPARATE phases. We train once on labeled data, then predict forever on new data.

Two Types of Supervised Learning

The difference comes down to ONE question: What kind of answer are we predicting?

REGRESSION

Predicts a NUMBER
The output is a continuous numerical value — it can be any number in a range.

Real-world examples:
- House price → $347,500
- Movie box-office revenue → $82M
- Temperature tomorrow → 24.3°C
- Patient's blood pressure → 128 mmHg
- Time to run a marathon → 3h 42m

Why useful: Any time we need a precise quantity, not just a category.

VS

CLASSIFICATION

Predicts a CATEGORY
The output is one of a fixed set of named categories — a discrete choice, not a number.

Real-world examples:
- Email → Spam OR Not Spam
- Tumour → Malignant OR Benign
- Iris flower → setosa / versicolor / virginica
- Customer → Will churn OR Stay
- Sentiment → Positive / Negative / Neutral

Why useful: Huge range of decision-making problems — is this safe? approved? fraud?

Regression — Predicting Numbers

Regression answers: "How much?" or "How many?" — the output is always a number.

From the slides — Movie Revenue Regression:

Input features: budget, genre, cast, director, release date, marketing spend… **Output:** Predicted box-office revenue ($)

Why use Regression instead of Classification here?

Classification would be WRONG here

If we said 'High / Medium / Low revenue' we LOSE precision.

$1M difference between predictions would matter to a studio. You can't capture that with a category.

Regression is CORRECT here

Regression gives the exact number: $82,450,000.

This precise value lets studios make real financial decisions — budgeting, marketing spend, theatre bookings.

Rule: If the output can be ANY value on a continuous scale → use Regression. If the output is one of N fixed categories → use Classification.

Classification — Predicting Categories

Classification answers: "Which group does this belong to?" — the output is always one of a fixed set of labels.

From the slides — Spam Email Classification:

Input: unlabeled emails (unknown if spam) **Training data:** thousands of emails already labeled spam / not spam **Output:** spam OR not spam

Two sub-types of Classification:

Binary Classification (2 classes)

Exactly 2 possible outputs:

- Spam / Not Spam
- Fraud / Not Fraud
- Malignant / Benign
- Win / Lose

Most common in practice. Simple and well-studied — many powerful algorithms.

Multi-class Classification (3+ classes)

3 or more possible outputs:

- Iris: setosa / versicolor / virginica
- Sentiment: Positive/Neutral/Negative
- Digit recognition: 0 through 9
- Disease type: A / B / C / D

Each data point gets exactly one label. The Iris dataset from these slides is multi-class.

How to Choose: Regression or Classification?

The type of your TARGET VARIABLE (what you're predicting) tells you which to use:

What TYPE is your target variable?

Is it a NUMBER?

→ Use REGRESSION

Is it a CATEGORY?

→ Use CLASSIFICATION

Problem	Target Variable	Type?	Algorithm
Predict house sale price	$247,000	Number → Continuous	REGRESSION
Classify email as spam	Spam / Not Spam	Category → 2 classes	CLASSIFICATION
Forecast tomorrow's temp	18.5°C	Number → Continuous	REGRESSION
Identify iris flower species	setosa / versicolor / virginica	Category → 3 classes	CLASSIFICATION
Predict patient survival days	127 days	Number → Continuous	REGRESSION

Machine Learning Vocabulary — The 4 Essential Terms

Every ML practitioner uses the same vocabulary. Master these 4 terms and you'll understand any ML paper or conversation:

TARGET

The column we want to predict — the 'answer' we're trying to learn.

Also called: output, label column, dependent variable, response variable, y

In the iris dataset → the TARGET is the 'species' column (setosa / versicolor / virginica)
Why: Without a target, we don't know WHAT to predict — it's the whole point of the exercise.

FEATURES

All other columns — the properties used as INPUT to make the prediction.

Also called: input variables, predictors, independent variables, X

In iris → FEATURES are sepal length, sepal width, petal length, petal width

Why: Features are the 'clues' the model uses. More relevant features = better predictions.

EXAMPLE

A single row of data — one individual data point in the dataset.

Also called: observation, instance, record, sample, data point

In iris → ONE EXAMPLE = one row: [6.7, 3.0, 5.2, 2.3, virginica]

Why: The model learns from thousands of examples — more examples usually means a better model.

LABEL

The target value for a SINGLE example — the answer for that one specific row.

Same as the target, but referring to one specific row rather than the whole column

In iris → the LABEL for the first row is 'virginica'

Why: Labels are what make supervised learning 'supervised' — the teacher's answers that the model learns from.

The Iris Dataset — A Classic ML Benchmark

The iris dataset is the 'Hello World' of machine learning — used in virtually every introductory ML course since 1936.

150	4	3	50
total examples (rows)	features (columns)	classes (species)	examples per class

Sample rows from the Iris dataset:

sepal length	sepal width	petal length	petal width	species ← TARGET
6.7	3.0	5.2	2.3	virginica
6.4	2.8	5.6	2.1	virginica
4.6	3.4	1.4	0.3	setosa
6.9	3.1	4.9	1.5	versicolor
4.4	2.9	1.4	0.2	setosa
5.9	3.0	5.1	1.8	virginica

← These 4 columns are the FEATURES (measurements in cm)

This column is the TARGET →

Vocabulary in Action — Identifying the TARGET

TARGET = The column we want to PREDICT. In iris: the 'species' column — setosa, versicolor, or virginica.

sepal length	sepal width	petal length	petal width	species ★ TARGET
6.7	3.0	5.2	2.3	**virginica**
6.4	2.8	5.6	2.1	**virginica**
4.6	3.4	1.4	0.3	**setosa**
6.9	3.1	4.9	1.5	**versicolor**
4.4	2.9	1.4	0.2	**setosa**

Why is species the TARGET?

Because that is what we want to PREDICT for new flowers. If we have a new iris flower and measure its petals and sepals, we want to know: is this setosa, versicolor, or virginica? The model learns from the species labels in the training data and then predicts species for flowers it has never seen before.

```
In Python:   X = df.drop('species', axis=1)    y = df['species']   → y is the target
```

Vocabulary in Action — Identifying the FEATURES

FEATURES = All the INPUT columns (non-target). In iris: sepal length, sepal width, petal length, petal width.

sepal length ★	sepal width ★	petal length ★	petal width ★	species
6.7	3.0	5.2	2.3	virginica
6.4	2.8	5.6	2.1	virginica
4.6	3.4	1.4	0.3	setosa
6.9	3.1	4.9	1.5	versicolor

Why do these 4 measurements work as features?

They are MEASURABLE — you can measure any iris flower the same way, numerically.

They are PREDICTIVE — different species have measurably different petal/sepal sizes.

They are GENERALIZABLE — the same measurements work for new unseen flowers.

If a feature did NOT vary between species, it would be USELESS — no predictive power.

Vocabulary in Action — EXAMPLE and LABEL

EXAMPLE = A single ROW in the dataset — one complete data point (all features + label together).

LABEL = The TARGET VALUE for ONE specific example — the correct answer for that single row.

sepal length	sepal width	petal length	petal width	species
6.7 ◀	3.0 ◀	5.2 ◀	2.3 ◀	virginica ★
6.4	2.8	5.6	2.1	virginica
4.6	3.4	1.4	0.3	setosa
6.9	3.1	4.9	1.5	versicolor

GREEN ROW = One EXAMPLE

The entire row [6.7, 3.0, 5.2, 2.3, virginica] is ONE example — one iris flower measured in the field. The dataset has 150 such examples.

STAR CELL = The LABEL

The label for this example is 'virginica' — that is the confirmed species identity for THIS specific flower. This is what the model is trained to predict.

All Four Vocabulary Terms — Side by Side

Term	What is it?	Other names	In Iris dataset	In Python (sklearn)
TARGET	The column to predict	output, y, label column	species	`y = df['species']`
FEATURES	Input columns (all non-target)	X, predictors, inputs	sepal/petal length & width	`X = df.drop('species',axis=1)`
EXAMPLE	One single row of data	observation, instance, row	[6.7, 3.0, 5.2, 2.3, virginica]	`X.iloc[0]` (first row)
LABEL	Target value for one example	answer, ground truth, tag	'virginica' (for row 1)	`y.iloc[0]` → `'virginica'`

How they relate to each other:

DATASET = many **EXAMPLES** | Each EXAMPLE has **FEATURES** + a **LABEL** | The **LABEL** column across ALL rows = the **TARGET**

Common Mistakes in Supervised Learning

Confusing Regression and Classification

Using a classifier when you need a number (or vice-versa). Always ask: is my target a number or a category?

Fix: Look at your target column. If it has infinite possible values → Regression. If it has a small fixed set of values → Classification.

Too little training data

A model trained on 10 examples of spam will fail on real email. You need enough labeled examples to cover the full range of variation.

Fix: Rule of thumb: at least 10× more examples than features. More is always better for supervised learning.

Using the target as a feature

Including the target column in the feature set (data leakage). The model would 'cheat' — perfectly predicting what it already knows.

Fix: Always drop the target column before passing features to the model. Use: X = df.drop('target', axis=1)

Non-representative training data

Training on data from one city and predicting for another. If patterns differ, the model will fail to generalize.

Fix: Ensure your training data comes from the same distribution as future prediction data.

Learning from mistakes is part of the process — even experienced data scientists make all of these!

Session Summary — Everything in One Place

01

Machine Learning learns from data

Instead of hand-coding rules, ML algorithms discover patterns automatically from labeled examples.

02

Supervised = labeled data

Every training example has a known correct answer (label). The model learns to predict these labels.

03

Regression predicts NUMBERS

Continuous output — house prices, temperatures, revenue. Output can be any value on a scale.

04

Classification predicts CATEGORIES

Discrete output — spam/not spam, species names, diagnosis. Output is one of N fixed labels.

05

4 essential vocabulary terms

Target (what to predict) · Features (inputs) · Example (one row) · Label (target for one example)

06

Pipeline: fit → predict

Train on labeled data with .fit(), then predict new labels on unseen data with .predict(). Two separate phases.

You now understand

Supervised Learning!

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